

LE MARS, IA, USA

WMO: 725484

Lat: 42.778N

Lon: 96.194W

Elev: 365

StdP: 97.02

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04942

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.1	-19.0	-26.0	0.4	-21.0	-22.9	0.5	-18.4	12.6	-6.3	11.5	-5.4	1.8	320	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	33.7	24.7	32.3	24.0	30.9	23.0	26.7	31.4	25.6	30.0	24.4	29.0	5.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	21.3	29.9	24.0	19.7	28.9	22.8	18.3	27.6	85.9	31.1	81.0	30.0	75.8	29.4	34.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.8	8.7	DB	-26.9	35.9	3.2	1.7	-29.2	37.1	-31.1	38.1	-32.9	39.0	-35.2	40.2	(4)
(5)			WB	-27.3	28.0	3.1	1.4	-29.5	29.0	-31.3	29.8	-33.0	30.6	-35.3	31.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	-6.1	-4.3	2.7	9.5	15.7	21.8	23.7	22.3	17.9	10.8	2.5	-4.1	(6)	
	DBStd	11.89	6.92	7.06	6.78	5.72	4.82	3.74	3.44	3.26	4.95	5.15	5.99	6.18	(7)	
	HDD10.0	1949	498	402	239	77	8	0	0	3	53	232	437		(8)	
	HDD18.3	3753	756	634	484	271	111	11	3	68	240	474	696		(9)	
	CDD10.0	1743	0	0	14	60	184	354	425	382	240	77	7	0	(10)	
	CDD18.3	506	0	0	1	4	28	115	170	128	54	5	0	0	(11)	
	CDH23.3	5224	0	0	9	74	343	1203	1810	1176	538	69	2	0	(12)	
	CDH26.7	1890	0	0	1	20	108	453	724	394	174	16	0	0	(13)	
(14) Wind	WSAvg	3.7	3.8	3.9	4.3	4.6	4.3	3.6	2.9	2.7	3.3	3.6	3.8	3.7	(14)	
(15) Precipitation	PrecAvg	609	13	14	42	63	84	95	72	80	72	43	26	16	(15)	
	PrecMax	928	50	44	123	191	227	204	214	317	226	130	101	58	(16)	
	PrecMin	281	0	0	0	10	13	19	13	9	14	0	0	0	(17)	
	PrecStd	137	11	11	30	41	36	50	43	57	45	33	24	13	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	15.1	24.9	29.1	32.5	35.1	36.1	34.9	33.6	28.9	22.1	12.7	(19)	
		MCWB	6.6	9.7	14.2	16.8	19.8	24.2	25.8	26.1	22.8	18.3	13.3	8.6	(20)	
	2%	DB	7.6	10.2	20.2	25.1	29.7	32.7	33.8	32.4	31.0	25.1	17.6	9.1	(21)	
		MCWB	4.1	6.5	12.7	15.1	19.4	23.4	25.5	25.5	21.9	15.6	11.4	6.0	(22)	
	5%	DB	4.9	7.3	16.3	22.4	27.4	31.2	32.3	30.3	28.3	22.5	14.1	6.3	(23)	
		MCWB	2.5	4.6	11.4	13.8	17.6	22.2	24.9	24.3	20.8	14.8	8.7	3.3	(24)	
	10%	DB	2.7	4.8	12.7	19.7	24.9	29.1	30.9	28.8	27.0	20.0	11.3	3.8	(25)	
		MCWB	1.3	2.8	8.5	12.4	16.5	21.2	24.0	23.4	20.1	13.7	7.3	2.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	11.1	16.9	19.0	23.5	26.7	29.6	28.3	25.1	21.0	14.9	10.0	(27)	
		MCDB	12.1	16.0	21.4	24.7	28.8	32.3	33.7	33.0	30.0	25.1	18.9	11.8	(28)	
	2%	WB	4.5	7.0	14.0	16.4	21.0	25.3	27.2	26.4	23.9	18.0	11.9	5.9	(29)	
		MCDB	7.3	9.6	18.3	23.3	26.9	30.6	31.8	31.1	28.4	22.9	17.3	8.7	(30)	
	5%	WB	2.8	4.6	11.6	14.8	19.2	24.0	25.9	25.1	22.7	16.1	9.7	3.9	(31)	
		MCDB	4.5	7.0	16.2	20.9	24.8	29.4	30.2	29.2	26.8	20.7	14.0	6.3	(32)	
	10%	WB	1.3	2.9	8.6	13.0	17.7	22.5	24.8	23.9	21.2	14.3	7.6	2.2	(33)	
		MCDB	2.9	4.7	13.4	18.5	22.8	27.6	29.1	28.0	25.3	18.7	10.6	3.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	9.9	11.0	12.6	12.1	11.5	11.5	11.2	13.0	12.7	11.2	9.4	(35)	
		MCDBR	13.1	12.9	17.4	17.3	15.8	13.1	12.2	12.4	14.6	17.0	16.0	12.9	(36)	
	5% WB	MCWBR	9.9	9.6	10.7	9.1	7.5	6.2	6.4	7.1	7.7	9.4	10.1	9.8	(37)	
		MCWBR	12.5	12.1	16.1	14.4	12.9	11.3	10.7	10.9	12.0	13.7	14.8	12.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.277	0.287	0.322	0.378	0.405	0.410	0.420	0.416	0.369	0.330	0.294	0.281	(40)	
		taud	2.425	2.420	2.405	2.264	2.264	2.311	2.278	2.278	2.380	2.468	2.490	2.435	(41)	
	Ebn at Noon		871	921	923	887	870	864	850	841	860	857	845	830	(42)	
		Edh at Noon	75	89	104	129	133	127	130	126	106	85	71	69	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.92	2.99	3.83	4.86	5.67	6.33	6.61	5.52	4.45	3.06	2.12	1.56	(44)	
	RadStd		0.17	0.24	0.30	0.40	0.42	0.46	0.32	0.40	0.30	0.37	0.13	0.13	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (47 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page